TED University
CMPE 491 Senior Design Project I

Project Proposal

Name of the Project	Attention-Guided Manipulation-Aware Detection of AI-Generated and Edited Images for Improved Cross-Dataset Generalization
Project Web Page	https://tunacantungul.github.io/manipulation-aware-detection/
Team Members	Ali Efe Altun Nilsu Dila Cabbar Tuna Can Tungul
Supervisor	Eren Ulu
Jury Members	Ayşe Yasemin Seydim Ayşenur Birtürk

Project Description

Detectors of AI-generated and AI-edited photographs often report near-perfect accuracy on the dataset they were trained on, yet degrade sharply on images from unseen generators, unseen editing tools, unseen datasets, or after real-world post-processing such as compression and resizing. A likely reason is that detectors learn shortcuts: cues that are correlated with the label in the training data but are not the manipulation itself.

This project investigates whether guiding a detector's attention toward the manipulated regions of an image improves its cross-dataset generalization. We add an attention module to an existing detector and supervise it with manipulation masks, which are available by construction for locally edited (inpainted) photographs. We evaluate the detector under cross-generator, cross-dataset and post-processing shifts, and measure whether its attention stays on the manipulated region when the data distribution changes.